Yuxing Liang

Ph.D. Candidate in Economics | Concordia University

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Education

Ph.D. in Economics, Concordia University, Montréal, Canada — Expected April 2027
Supervisor: Szilvia Pápai
External research advisor: Xintong Han
M.A. in Economics, Concordia University, Montréal, Canada — 2018–2020
B.A. in Economics, Shanghai Customs College, Shanghai, China — 2014–2018

Research Profile

Fields: Market Design, Matching Theory, Applied Microeconomic Theory
Research interests: Strategy-proofness, preference elicitation, assignment mechanisms, school choice, college and major assignment
Methods: Computational analysis, simulation, empirical analysis

Job Market Paper

False but Harmless? Misreporting under Strategy-Proof Matching

Strategy-proof matching mechanisms make truth-telling a weakly dominant strategy, yet many students submit nontruthful rankings. A misreport costs nothing if it gives the student the same school as the truth, whatever the others report. This paper characterizes these costless misreports when every student must rank every school. Under Deferred Acceptance, Top Trading Cycles, and every strategy-proof mechanism that respects top-priority sets, is top-top consistent, and grants feasible first choices, each student has a guarantee rank, computed from her own ranking, the priorities, and the capacities. Reordering the schools below this rank never costs her anything. With at least one more seat than students, the costless misreports are exactly those that reorder only the schools below the guarantee rank, in every priority profile and under all these mechanisms; without a spare seat, some priority profiles allow other costless misreports. The results give an explanation of misreporting that complements behavioral accounts, and show which part of a submitted ranking reveals a student's preferences. In the Chen–Sönmez experiment, 38 of 58 nontruthful rankings kept the realized assignment, but only 6 are dominant; the other 52 need a different explanation.

Working Papers

Towards a Better Major Assignment: An Empirical Study with Students in China (with Xintong Han and Yan Zeng), *submitted*

Using students' submitted rankings, exam records, and major characteristics from a Chinese university, we estimate preferences and compare assignment reforms through counterfactual analysis. The results show how alternative mechanisms affect welfare, first-choice placement, and the distributional consequences of preferential treatment.

Progressive Choice Mechanisms in School Choice (with Szilvia Pápai)

We study a family of mechanisms that varies how many schools each student may consider at each step, connecting Immediate Acceptance, Chinese parallel mechanisms, Serial Dictatorship, and Deferred Acceptance. We establish efficiency under common priorities, show that broader choice opportunities weakly reduce manipulability, and study the corresponding stability trade-offs.

Work in Progress

Bridging Theory and Evidence: How External Information Transforms Students' Application Decisions

This project studies how feedback about relative rank changes students' stated preferences and applications. It connects information acquisition to assignment design by tracing how students revise the comparisons that shape their submitted rankings.

Selected Presentations

False but Harmless? Misreporting under Strategy-Proof Matching. 18th Meeting of the Society for Social Choice and Welfare, University of Tokyo — June 2026
False but Harmless? Misreporting under Strategy-Proof Matching. 20th CIREQ–CIRANO Ph.D. Students' Conference, CIRANO — May 2026
False but Harmless? Misreporting under Strategy-Proof Matching. CIREQ Interdisciplinary Ph.D. Conference in Applied Market Design, Concordia University — March 2026
Progressive Choice Mechanisms in School Choice. Management Science Workshop Series (virtual), Universidad Católica de Chile — October 2025
Progressive Choice Mechanisms in School Choice. 14th Conference on Economic Design, University of Essex — June 2025
Progressive Choice Mechanisms in School Choice. 17th Meeting of the Society for Social Choice and Welfare, Paris School of Economics — July 2024
Towards a Better Major Assignment: An Empirical Study with Students in China. Economics Department Lunch Seminar, Concordia University — November 2023
Towards a Better Major Assignment: An Empirical Study with Students in China. Urban and Education Economics Seminars, Shandong University — July 2023
Towards a Better Major Assignment: An Empirical Study with Students in China. 13th Conference on Economic Design, Universitat de Girona — June 2023
Towards a Better Major Assignment: An Empirical Study with Students in China. 18th CIREQ Ph.D. Students' Conference, Concordia University — May 2023
Progressive Choice Mechanisms in School Choice. Agents' Behavior in Combinatorial Game Theory, CRM Montréal (virtual) — November 2021

Course Instructor, Concordia University

- **ECON 332** Introductory Financial Economics, Winter 2027 (scheduled)
- **ECON 436/536** Economics of Taxation, Fall 2026
- **ECON 491/591** Environmental Economics, Winter 2026
- **ECON 221** Statistical Methods I, Winter 2025
- **ECON 377** The Asia–Pacific Rim Economies, Winter 2024

Tutorials and Teaching Assistantships, Concordia University

Tutorial Leader

- **ECON 612** Microeconomics I, Winter 2023

Teaching Assistant

Graduate courses

- **ECON 614** Game Theory, Winter 2022 and Fall 2025
- **ECON 642** Financial Economics I, Fall 2024
- **ECON 695** Empirical Industrial Organization, Fall 2023
- **ECON 612** Microeconomics I, Winter 2023
- **ECON 673** Industrial Organization, Fall 2022

Undergraduate courses

- **ECON 221** Statistical Methods I, Fall 2025
- **ECON 301** Intermediate Microeconomic Theory I, Fall 2023
- **ECON 485** Health Economics, Winter and Fall 2023
- **ECON 461/561** Industrial Organization, Winter 2023
- **ECON 464** Game Theory, Fall 2022
- **ECON 422** Econometrics II, Winter 2022
- **ECON 203** Macroeconomics, 2020–2025
- **ECON 201** Microeconomics, 2020–2025

Research Experience

Research Assistant, Concordia University — Winter 2023

Prepared research datasets, conducted literature reviews, and supported statistical analysis for faculty research.

Research Assistant, Shanghai Customs College — 2017–2018

Collected and cleaned research data, reviewed related literature, and supported descriptive empirical analysis.

Computational Methods for Matching Research

Mechanism implementation and simulation — *MATLAB, Python*

Implement and compare Serial Dictatorship, Immediate Acceptance, Deferred Acceptance, Top Trading Cycles, and Progressive Choice mechanisms; simulate finite markets and diagnose allocation outcomes.

Preference and assignment data analysis — *Python, Stata, R*

Construct reproducible rank-order-list and assignment datasets, estimate discrete-choice models, and evaluate mechanism-based counterfactuals.

Computational support for theory — *Python, MATLAB*

Enumerate finite markets, search for counterexamples, stress-test conjectures, and verify theoretical boundaries across market environments.

Selected Grants and Awards

FRQSC Doctoral Research Scholarship (B2Z), 1st place, Fonds de recherche du Québec	*CAD $58,333*	2023–2026
Graduate Fellowship, Concordia University	*CAD $56,000*	2020–2024
International Tuition Award of Excellence, Concordia University	*CAD $40,656*	2020–2024
Professor Jaleel Ahmad Scholarship, Concordia University	*CAD $2,500*	2021–2022
SGS Conference and Exposition Allowance, Concordia University	*CAD $2,000*	2023, 2024
Research Bursary, CIREQ	*CAD $500*	2023
Conference Travel Funding, Faculty of Arts and Science, Concordia University	*CAD $400*	2023
Conference Travel Funding, Department of Economics, Concordia University	*CAD $400*	2023, 2024
Outstanding Undergraduate Thesis, Shanghai Customs College		2018

Additional Professional Experience

Analyst, KYC Periodic Team, KPMG Advisory (China) Limited, Shanghai — Jan. 2017–Feb. 2018
Market Research Assistant, Tebon Securities Co., Ltd., Shanghai — Aug. 2016–Jan. 2017

Technical Skills and Languages

Programming and analysis: Python, MATLAB, Stata, R, EViews, SAS; PostgreSQL, MySQL
Research workflow: LaTeX, GitHub, Microsoft Office, Power BI, Tableau
Languages: Mandarin (native), English (fluent), Japanese (intermediate), French (beginner)

References

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